Fall 2026 Exploratory Data Analysis with R

Lecture 2: Data Manipulation and the Tidyverse

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01

Recap & Agenda

Quick Recap: Last Week

- Got Posit Cloud set up and wrote our first lines of R
- Imported a dataset with `read_csv()`
- Took a first look at `unemp_df`: the U.S. unemployment rate, 1948–2022
- Got a first taste of `summary()` and `filter()`

Today: turn that first taste into real fluency.

Goals for Today

- Meet the pipe, `%>%`
- Learn the five core **dplyr** verbs: `select()`, `filter()`, `arrange()`, `mutate()`, `summarize()`
- Pair `group_by()` with `summarize()` to answer questions *by group*
- Reshape data from wide to long with `pivot_longer()`
- Apply all of it to `unemp_df` — and to the population data from Homework 1

02

The Pipe & Core Verbs

Reading Code Left to Right

Nested function calls get hard to read fast:

```r
arrange(filter(unemp_df, Year > 2000), desc(Unrate))
```

```
# A tibble: 22 × 2
    Year Unrate
   <dbl>  <dbl>
 1  2010   9.61
 2  2009   9.28
 3  2011   8.93
 4  2020   8.09
 5  2012   8.07
 6  2013   7.36
 7  2014   6.16
 8  2003   5.99
 9  2008   5.8
10  2002   5.78
# ℹ 12 more rows
```

To read this you have to work from the inside out. The **pipe operator**, `%>%`, lets us write the same thing as a sequence of steps, read top to bottom:

```
unemp_df %>%
  filter(Year > 2000) %>%
  arrange(desc(Unrate))
```

```
# A tibble: 22 × 2
    Year Unrate
   <dbl>  <dbl>
 1  2010   9.61
 2  2009   9.28
 3  2011   8.93
 4  2020   8.09
 5  2012   8.07
 6  2013   7.36
 7  2014   6.16
 8  2003   5.99
 9  2008   5.8
10  2002   5.78
# ℹ 12 more rows
```

How to Read `%>%`

- Read it as **“and then”**
- `x %>% f()` is exactly the same as `f(x)`
- `x %>% f() %>% g()` is exactly the same as `g(f(x))`

Every function in the **tidyverse** is built to work this way: take data in as the first argument, hand data back out, ready for the next step.

Choosing Columns: `select()`

Sometimes a dataset has more columns than we need. `select()` keeps only the ones we ask for.

```
unemp_df %>%
  select(Year, Unrate) %>%
  head(4)
```

```
# A tibble: 4 × 2
   Year Unrate
  <dbl>  <dbl>
1  1948   3.75
2  1949   6.05
3  1950   5.21
4  1951   3.28
```

`unemp_df` only has two columns already, so this one won’t look like much yet — but wait until we bring in a dataset with more of them.

`select()` Tricks

- `select(df, -Year)` — keep everything *except* `Year`
- `select(df, starts_with("Un"))` — keep columns whose names start with `"Un"`
- Column order in, column order out: `select()` also **reorders**

Exercise 1: Filtering, Properly

Last week we used `filter()` once. Let's push it further.

1. Filter `unemp_df` to years **after** 2007 (the leadup to the financial crisis)
2. Filter to years where `Unrate` was **above 6**
3. Combine both conditions into a single `filter()` call

Combining Conditions

```
unemp_df %>%
  filter(Year > 2007 & Unrate > 6)
```

```
# A tibble: 7 × 2
   Year Unrate
  <dbl>  <dbl>
1  2009   9.28
2  2010   9.61
3  2011   8.93
4  2012   8.07
5  2013   7.36
6  2014   6.16
7  2020   8.09
```

- `&` means **and** — both conditions must hold
- `|` means **or** — either condition can hold
- `%in%` checks membership: `Year %in% c(2008, 2009, 2020)`

Putting Things in Order: `arrange()`

```
unemp_df %>%
  arrange(Unrate) %>%
  head(5)
```

```
# A tibble: 5 × 2
   Year Unrate
  <dbl>  <dbl>
1  1953   2.92
2  1952   3.02
3  1951   3.28
4  1969   3.49
5  1968   3.56
```

That's the five **lowest**-unemployment years on record in this dataset. To flip the order:

```r
unemp_df %>%
  arrange(desc(Unrate)) %>%
  head(5)
```

```
# A tibble: 5 × 2
   Year Unrate
  <dbl>  <dbl>
1  1982   9.71
2  2010   9.61
3  1983   9.6
4  2009   9.28
5  2011   8.93
```

`arrange()` + `filter()` Together

The real power shows up once you start chaining verbs:

```
unemp_df %>%
  filter(Year >= 2000) %>%
  arrange(desc(Unrate))
```

```
# A tibble: 23 × 2
    Year Unrate
   <dbl>  <dbl>
 1  2010   9.61
 2  2009   9.28
 3  2011   8.93
 4  2020   8.09
 5  2012   8.07
 6  2013   7.36
 7  2014   6.16
 8  2003   5.99
 9  2008   5.8
10  2002   5.78
# ℹ 13 more rows
```

This reads as: *start with `unemp_df`, keep years from 2000 on, then sort by unemployment, highest first.*

Breakout Session 1

Task: Using `unemp_df` and the pipe, answer:

1. What are the three years with the **lowest** unemployment since 1990?
2. Was unemployment ever above 9% before the year 2000? Which years?
3. Sort all years from 1980–1989 by unemployment, worst to best.

Hint: every one of these is a `filter()` and/or `arrange()`, chained with `%>%`.

03

Creating New Variables

Creating New Columns: `mutate()`

Every verb so far has worked with columns that already exist. `mutate()` **creates** new ones.

```
unemp_df %>%
  mutate(decade = (Year %/% 10) * 10) %>%
  head(4)
```

```
# A tibble: 4 × 3
   Year Unrate decade
  <dbl>  <dbl>  <dbl>
1  1948   3.75   1940
2  1949   6.05   1940
3  1950   5.21   1950
4  1951   3.28   1950
```

`%/%` is integer division — `1987 %/% 10 = 198`, times 10 gives `1980`. This is a common trick for bucketing years into decades.

`mutate()` with `lag()`

A very common use case: comparing a row to the row *before* it.

```
unemp_df %>%
  mutate(change = Unrate - lag(Unrate)) %>%
  filter(Year >= 2007 & Year <= 2010)
```

```
# A tibble: 4 × 3
   Year Unrate  change
  <dbl>  <dbl>   <dbl>
1  2007   4.62 0.00833
2  2008   5.8  1.18
3  2009   9.28 3.48
4  2010   9.61 0.325
```

`lag(Unrate)` shifts the whole column down by one row, so `change` is the year-over-year move in the unemployment rate. Watch what happens around 2009.

Take Ten

Back at the top of the hour.

(Refill your coffee. Stretch. Do not open Canvas.)

04

Summarizing by Group

`group_by()` + `summarize()`

The single most useful pattern in the tidyverse: split your data into groups, then compute something for each group.

```r
unemp_df %>%
  mutate(decade = (Year %/% 10) * 10) %>%
  group_by(decade) %>%
  summarize(avg_unrate = mean(Unrate),
            max_unrate = max(Unrate))
```

`group_by()` doesn't change what the data looks like — it changes what happens **next**.

`summarize()` then collapses each group down to one row.

```
# A tibble: 9 × 3
  decade avg_unrate max_unrate
   <dbl>      <dbl>      <dbl>
1   1940       4.9        6.05
2   1950       4.51       6.84
3   1960       4.78       6.69
4   1970       6.22       8.48
5   1980       7.27       9.71
6   1990       5.76       7.49
7   2000       5.54       9.28
8   2010       6.22       9.61
9   2020       5.71       8.09
```

Exercise 2: By Decade

Using the `decade` column from `mutate()`:

1. Find the decade with the **highest average** unemployment
2. Find the single **worst year** within that decade
3. Do it all in one piped chain — no saving intermediate objects

Breakout Session 2

Work with a partner.

Task:

1. Compute average unemployment for every decade from 1950 through 2019
2. Sort the result from highest to lowest average
3. Which decade would you *least* want to be graduating into? Why?

Stretch goal: instead of `mean()`, try `sd()` — which decade had the most *volatile* unemployment, not just the highest?

05

Reshaping Data

Wide vs. Long Data

Remember `decade_pop.csv` from Homework 1? Let's bring it back.

```
pop_df
```

```
# A tibble: 24 × 4
    Year   Black    White Source
   <dbl>   <dbl>    <dbl> <chr>
 1  1790  757208  3172006 Census https://www.census.gov/content/dam/Census/libr…
 2  1800 1002037  4306446 Census https://www.census.gov/content/dam/Census/libr…
 3  1810 1377808  5862073 Census https://www.census.gov/content/dam/Census/libr…
 4  1820 1771656  7866797 Census https://www.census.gov/content/dam/Census/libr…
 5  1830 2328642 10532060 Census https://www.census.gov/content/dam/Census/libr…
 6  1840 2873648 14189705 Census https://www.census.gov/content/dam/Census/libr…
 7  1850 3638808 19553068 Census https://www.census.gov/content/dam/Census/libr…
 8  1860 4441830 26922537 Census https://www.census.gov/content/dam/Census/libr…
 9  1870 4880009 33589377 Census https://www.census.gov/content/dam/Census/libr…
10  1880 6580793 43402970 Census https://www.census.gov/content/dam/Census/libr…
# ℹ 14 more rows
```

This is **wide** data: each *row* is a decade, and `Black` and `White` are separate *columns*. It's easy to read, but awkward for the tidyverse — `ggplot()` and `group_by()` both want one **variable per column**, and here, "population" is really one variable split across two columns.

What We Want Instead

Long data would look like this:

Year	race	population
1790	Black	757208
1790	White	3172006
1800	Black	1002037
…	…	…

One row per Year–race combination. This is what `pivot_longer()` is for.

pivot_longer()

```
pop_long <- pop_df %>%
  pivot_longer(cols = c(Black, White),
               names_to = "race",
               values_to = "population")

pop_long %>% head(6)
```

- `cols` — which columns to fold together
- `names_to` — the new column that holds the old column *names* (`"Black"`, `"White"`)
- `values_to` — the new column that holds the values that used to live in those columns

```
# A tibble: 6 × 4
   Year Source                                          race  population
  <dbl> <chr>                                           <chr>      <dbl>
1  1790 Census https://www.census.gov/content/dam/Census/libra… Black     757208
2  1790 Census https://www.census.gov/content/dam/Census/libra… White    3172006
3  1800 Census https://www.census.gov/content/dam/Census/libra… Black    1002037
4  1800 Census https://www.census.gov/content/dam/Census/libra… White    4306446
5  1810 Census https://www.census.gov/content/dam/Census/libra… Black    1377808
6  1810 Census https://www.census.gov/content/dam/Census/libra… White    5862073
```

Now `group_by()` Actually Works

```
pop_long %>%
  group_by(race) %>%
  summarize(pop_1790 = min(population),
            pop_2020 = max(population),
            growth = max(population) - min(population))
```

```
# A tibble: 2 × 4
  race   pop_1790  pop_2020    growth
  <chr>     <dbl>     <dbl>     <dbl>
1 Black    757208  41100000  40342792
2 White   3172006 223553265 220381259
```

That’s the whole point of reshaping: once “race” is a column instead of two separate columns, `group_by(race)` is one line of code instead of a headache.

Breakout Session 3

Using `pop_long` from the last section:

1. Compute each group's **share** of the total population in 1900, 1950, and 2000 (`filter()` + `mutate()`)
2. Which decade had the fastest growth for each group? (Hint: you'll want `arrange()` inside each group — look up `group_by() %>% arrange()`)
3. In your own words: why was reshaping to long format a **necessary** step before you could answer question 2?

06

Agentic Coding

What Is Agentic Coding?

So far, every line of R in this class has been typed by a person. **Agentic coding** means prompting an LLM — Claude, in these examples — in plain English, and letting it write the code for you.

This is quickly becoming a normal part of professional data work, including at the Fed. It doesn't replace the tidyverse verbs from today — it depends on them.

Prompting Claude for a FRED Pull

In Homework 1 you set up a FRED account, requested an API key, and used `fredr_set_key()` to connect to it. Assume that's already done. Here's a prompt you might give Claude:

> I already have a FRED API key set with `fredr_set_key()`. Using the `fredr` and `dplyr` packages, write R code that pulls the monthly `UNRATE` series back to 1990, buckets the observations by decade, and reports the average unemployment rate for each decade, sorted from highest to lowest.

Claude might hand back something like this:

```r
library(fredr)
library(tidyverse)
library(lubridate)

unrate <- fredr(
  series_id = "UNRATE",
  observation_start = as.Date("1990-01-01")
)

unrate %>%
  mutate(decade = (year(date) %/% 10) * 10) %>%
  group_by(decade) %>%
  summarize(avg_unrate = mean(value, na.rm = TRUE)) %>%
  arrange(desc(avg_unrate))
```

Notice the shape: `mutate()`, `group_by()`, `summarize()`, `arrange()`. Four verbs from today's lecture, chained together. The agent didn't invent a new way to answer the question — it used the same toolkit you're building.

Checking the Agent's Work

`fredr` needs a live connection and your personal key, so we can't run Claude's exact code here. Notice, too, that the column names are different from what we're used to (`date` and `value`, not `Year` and `Unrate`) — that alone is worth catching before you trust the output.

We can still check the *logic* against data we already have loaded:

```
unemp_df %>%
  filter(Year >= 1990) %>%
  mutate(decade = (Year %/% 10) * 10) %>%
  group_by(decade) %>%
  summarize(avg_unrate = mean(Unrate)) %>%
  arrange(desc(avg_unrate))
```

```
# A tibble: 4 × 2
  decade avg_unrate
   <dbl>      <dbl>
1   2010       6.22
2   1990       5.76
3   2020       5.71
4   2000       5.54
```

Same question, same answer shape. The point isn't that the numbers match exactly — FRED revises data, and today's date keeps moving — it's that you can independently reproduce the claim with verbs you already own, instead of taking it on faith.

Exercise 3: Prompt, Then Verify

1. Ask an LLM for R code, using `fredr`, that finds which decade since 1990 had the most **volatile** unemployment rate — not the highest average, the highest standard deviation.
2. Run the code it gives you.
3. Using only `mutate()`, `group_by()`, and `summarize()` on `unemp_df`, reproduce the same calculation yourself.
4. If the two don't agree, figure out why before you trust either answer.

Aesthetic Agents

Agents don’t just get the logic right — they have opinions about how a chart should *look*. Reuse the setup from before, but this time ask for a figure:

> Same FRED setup as before — my key is set with `fredr_set_key()`. Using `fredr`, `dplyr`, and `ggplot2`, pull monthly `UNRATE` since 1990 and plot it over time. Keep it **clean and free of visual distractions**, with proper axis labels, a clear title, and the data source as a **left-aligned caption**.

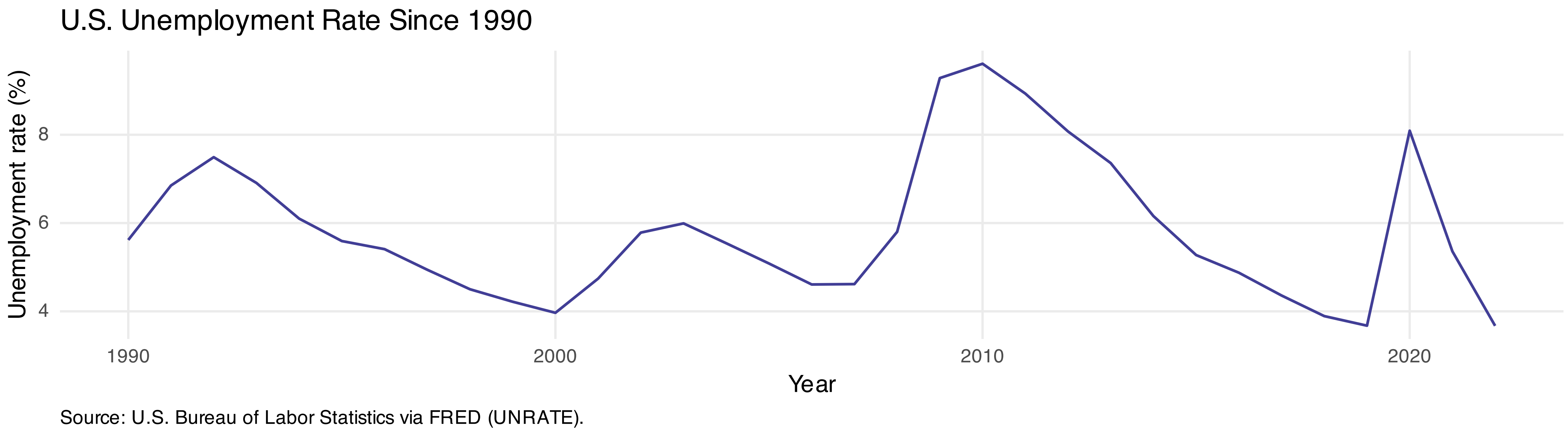
Claude might hand back:

```r
library(fredr)
library(tidyverse)

unrate <- fredr(
  series_id = "UNRATE",
  observation_start = as.Date("1990-01-01")
)

ggplot(unrate, aes(x = date, y = value)) +
  geom_line(color = "#413e96", linewidth = 0.7) +
  labs(
    title   = "U.S. Unemployment Rate Since 1990",
    x       = "Year",
    y       = "Unemployment rate (%)",
    caption = "Source: U.S. Bureau of Labor Statistics via FRED (UNRATE)."
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.caption      = element_text(hjust = 0),
    panel.grid.minor  = element_blank()
  )
```

Every choice is deliberate: `theme_minimal()` drops the gray panel and heavy gridlines, the minor grid is switched off, both axes are labeled in plain English, and `hjust = 0` pushes the source note to the left margin.

The Agent's Eye for Design

We can't run the `fredr` pull live, but the same styling drawn from `unemp_df` produces this:



Clean lines, honest labels, no chartjunk — the aesthetic defaults an agent reaches for are usually the ones a careful analyst would choose anyway.

Ground Rules for Agentic Coding

- **Never paste a real key or password into a prompt.** Treat a chat window like any other place code you didn't write yet runs. Use a placeholder like `fredr_set_key("YOURKEYHERE")`, exactly as Homework 1 does.
- **Read the code before you run it.** A confident-sounding agent can still call a function that doesn't exist, or reach for the wrong column name.
- **Verify with tools you understand.** Today's `mutate()`, `group_by()`, and `summarize()` are enough to catch most mistakes.
- **Save your progress as a markdown file!** Don't lose all your hard work coming up with visually-appealing charts. Prompt your agent to save what it learned as a markdown file so you can carry it with you to other projects or LLMs.
- Agentic coding multiplies the verbs you already have. It isn't a substitute for learning them.

07

Recap

Recap

Today we added five verbs to the toolkit:

- `select()` — choose columns
- `filter()` — choose rows
- `arrange()` — reorder rows
- `mutate()` — create columns
- `group_by()` + `summarize()` — compute *by group*

And one reshaping tool: `pivot_longer()`, for turning wide data into tidy, analysis-ready long data.

Chained together with `%>%`, these five verbs cover the overwhelming majority of day-to-day data wrangling — in this class, and afterward.

Links & Further Reading

- R for Data Science, Ch. 5: Data Transformation
- RStudio dplyr cheat sheet
- tidyr pivoting vignette

Coming up: more summary statistics, and starting the project.